Chapter 5

AI in Research, Generative AI and Other Important Issues

In the constantly evolving realm of scientific investigation, the integration of Artificial Intelligence (AI) stands out as a transformative influence, fundamentally altering the approach researchers take toward specific challenges. This chapter explores the convergence of AI and scientific research, delving into the dynamic contributions of intelligent algorithms as they contribute to the pursuit of knowledge. Additionally, the emergence of Generative AI has brought forth numerous possibilities and applications for AI. Within this chapter, there will be a broad introduction to Generative AI, accompanied by an overview of its most prominent product, namely ChatGPT. Notably, Prompt Engineering is highlighted as a crucial skill, poised to become the next significant expertise, as it plays a pivotal role in unlocking the true capabilities of ChatGPT. The narrative extends to include a discussion on other noteworthy advancements in the field of AI.

5.1 AI IN EXPERIMENTATION AND MULTI-DISCIPLINARY RESEARCH

There are many activities in experimentation and research where AI can be of help. Some indicative steps are the Generation of Hypothesis, Virtual Prototyping or Simulation, Collection of the data and then doing analysis or finding patterns. In the following section, we will explore the use of AI in various disciplines and sub-disciplines.

5.1.1 Application in Particle Physics (Identifying Particles in Collider Data)

At the Large Hadron Collider (LHC), AI algorithms are used to shift through massive amounts of data and accurately identify different parti-

cles, such as Higgs bosons, among billions of collisions. Located in Geneva, Switzerland this is definitely one of the wonders of Science and Engineering. Designed to collide protons and ions at close to the speed of light, recreating conditions similar to the Big Bang. This task would be extremely time-consuming for humans to do manually. Thousands of scientists from over 100 countries work on the LHC (**Fig. 5.1**).

CERN runs highly complex scientific instruments that require advanced control systems and produce enormous amounts of research data. Managing these complex systems and analysing massive datasets pose major challenges.



Figure 5.1 LHC Tunnel of CERN

To address these challenges, CERN is leveraging artificial intelligence techniques in various ways. AI is helping CERN improve particle beam handling through smarter control algorithms. It enables more efficient analysis of the petabyte-scale data generated by CERN's experiments. And AI techniques are being applied to optimize maintenance and upkeep of CERN's large-scale facilities. By harnessing artificial intelligence, CERN aims to enhance many aspects of operating its sophisticated scientific infrastructure for particle physics research.

5.1.2 Application in Astro Physics and Space

AI is used to analyse images of galaxies, stars, and other celestial objects to detect patterns, classify objects, and identify anomalies that might lead to new discoveries. Machine learning models can simulate the evolution of galaxies, predict the behaviours of stars, or simulate the effects of gravitational interactions. AI is used to simulate complex astrophysical phenomena, such as galaxy formation, star formation, and the dynamics of celestial bodies. This helps researchers understand the underlying processes and test theoretical models.

Scientists studying space have had a problem when creating simulations —they had to choose between making detailed simulations or covering a lot of the universe. Now, with the help of a technology called generative adversarial networks, they can do both.

Researchers from Carnegie Mellon University and the University of California came up with a smart computer program. This program can take basic simulations and turn them into highly detailed ones quickly. So now, scientists can make complex simulated universes in just one day. AI is also being used for mapping of dark matters (**Fig. 5.2**)



Figure 5.2 Mapping of Dark Matters using AI

5.1.3 AI in Experimental Chemistry

AI can assist in various aspects of experimental chemistry, offering benefits such as improved efficiency, data analysis and the discovery of novel compounds. The AI algorithms can analyse data from previous experiments to suggest optimal conditions for reactions, helping researchers design experiments with higher success rates. AI can predict reaction pathways and propose optimal sequences of reactions, facilitating the planning of complex syntheses. The AI algorithms can analyse spectroscopic data (e.g., NMR, IR, UV-Vis) to identify compounds, determine reaction progress, and characterize reaction products. It can help, process and analyse large datasets of chemical information, facilitating the extraction of meaningful patterns and insights. AI assists in the virtual screening of potential drug candidates, predicting their binding affinities to specific biological targets and prioritizing compounds for synthesis and testing.

Revolutionizing the art of molecule-making, AI retrosynthesis tools are rewriting the rules of chemical synthesis. These digital wizards take a target molecule as their canvas, then paint in reverse, meticulously charting the ideal pathway back to its simplest building blocks. One promising approach, 3N-MCTS, blends a timeless search algorithm with the power of three neural networks, unveiling the most efficient route from concept to creation. Despite the buzz these tools generate, the majority of chemists are yet to fully embrace their transformative potential.

5.1.4 AI in Biology

AI algorithms are deciphering the vast libraries of genetic information, identifying patterns and connections that would elude even the most dedicated human researchers. This is leading to breakthroughs in understanding diseases, personalized medicine, and even the evolution of life itself. Some of the use cases of AI in biology are as follows:

- Predicting protein structures from amino acid sequences using deep learning.
- Identifying disease risk factors and biomarkers from genetic data through machine learning.

- Automating the analysis of microscope images to detect cells, tissues, etc., via computer vision.
- Discovering new drugs and materials by screening virtual libraries with AI algorithms.
- Modelling complex biological systems like gene networks and cell interactions using Bayesian networks and machine learning.
- Synthesizing novel proteins with desired functions using generative AI models.
- Classifying bacteria species rapidly using image recognition and spectral data with neural networks.
- Tracking neural activity in the brain in real-time using AI signal processing and analytics.
- Mining scientific literature to extract biological knowledge and relationships using natural language processing.

One of the leading Research Organizations shows (**Fig. 5.3**) how Global AI Genomics is poised to grow.

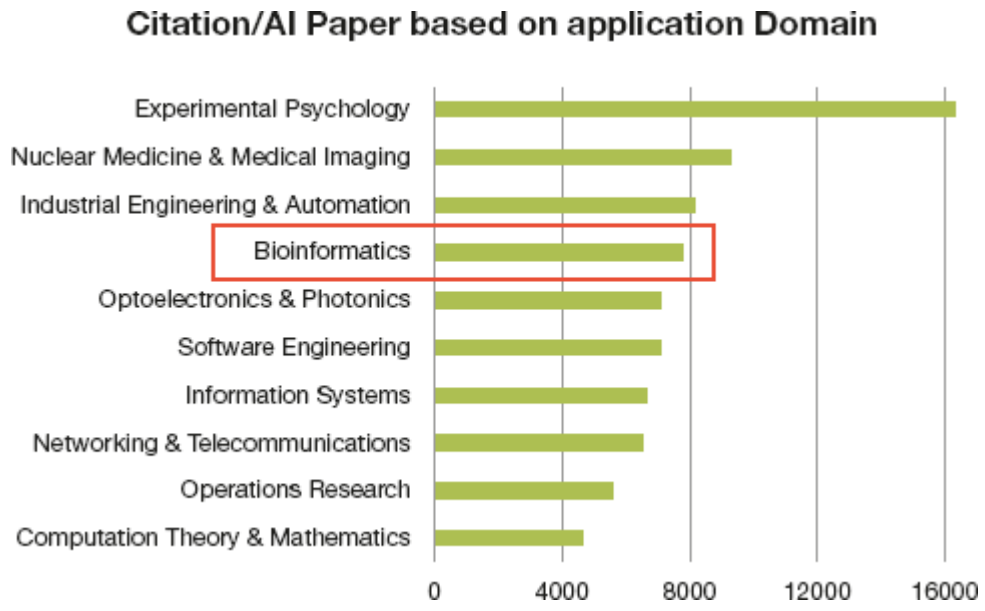


Figure 5.3 Growth Projection of Genomics Market in the US

5.1.5 AI in Environmental Science

Artificial Intelligence (AI) can be applied in various ways in environmental science to address challenges, improve efficiency and enhance decision-making.

- **Climate Modelling and Prediction:** AI can analyse vast amounts of climate data to improve climate models and enhance predictions. Machine learning algorithms can identify patterns, trends, and correlations in climate data, leading to more accurate and timely climate forecasts.
- **Air and Water Quality Monitoring:** AI technologies can be used to analyse sensor data and satellite imagery to monitor air and water quality. Machine learning algorithms can identify patterns and anomalies, providing insights into pollution sources, potential health risks, and environmental impacts.
- **Waste Management:** AI technologies, including robotics and machine learning, can be applied to improve waste sorting, recycling processes and waste management overall. Automated systems can identify recyclable materials and help reduce the environmental impact of waste disposal.
- **Resource Management and Conservation:** AI algorithms can optimize resource management strategies, such as precision agriculture. By analysing data on soil conditions, weather patterns and crop health, AI can help farmers make more informed decisions, reduce resource use, and minimize environmental impact.



POINTS TO PONDER

AI for Penguins

In 2017, there was a very interesting competition launched by DrivenData. Antarctic penguins are hugely popular and appealing animals known for their distinctive behaviour. Their quirky nature has been documented extensively by wildlife photographers and researchers. To better manage and conserve penguin species over the long term, accurately modelling and predicting their population numbers is essential. It also provides insights into the dynamics of the broader Antarctic ecosystem. This data science competition aimed to leverage the skills of the data science community to develop models that can effectively forecast penguin population changes. By tapping into innovative modelling approach-

es, the goal was to produce more accurate and robust predictions of penguin populations and their shifts. Reliable models for anticipating population trends are vital tools for enacting evidence-based conservation policies for Antarctic penguin species. One of the prominent species is Emperor Penguin (**Fig. 5.4**).



Figure 5.4 Herd of Emperor Penguin

5.2 GENERATIVE AI INTRODUCTION

If we look at this topic from just a semantics perspective, generative AI is a specialized branch of AI, which can generate new content. That content may be new text, that may be images, videos, songs and of course a mixture of the same. Specifying that it can generate is not enough. The content should be such that, it is coherent and must pass the 'Turing Test', i.e., it should not appear as machine-generated. Predominantly, deep learning or neural network-based models are used as generative models.

Machine learning or deep learning models can be broadly classified into (a) Discriminative Models and (b) Generative Models. Enlisted are some of the key points about these models.

1. **Discriminative Model:**
    1. Models' conditional probability $p(y \mid x)$ of outputs y given inputs x
    2. Focuses on predicting y from x, not the full input distribution
    3. **Examples:** Logistic Regression, SVMs, Neural Networks
    4. Does not generate new data points
    5. Better for classification and prediction tasks
2. **Generative Model:**
    1. These models the joint probability $p(x, y)$ of inputs x and outputs y

2. Learns the distribution of all input data
3. **Examples:** Naive Bayes, LDA, Gaussian Mixture Models
4. Can generate or sample new data points
5. Good for modelling input variability

In a layman's term, given different attributes of an unknown species, discriminative model will have the capability to label it to one of the known categories. It will say this looks more like a dog, than a cat. The generative model will try to understand what makes a dog a dog. Try to unearth the underlying probability distribution behind the same. As a result, it will develop the capability of generating new dogs. I think, it is not difficult to understand that the goal of the generative model is much more lofty.

The current generative models are backed by a huge amount of data. They are gigantic models with billions of parameters. Many big organizations were working on building generative models.

5.2.1 Chronology of the Different Developments

However, there is no denying that OpenAI was the first to take the world by storm and showed what is the capability of a generative AI model in a democratized fashion. Soon other giants like Meta and Google joined the bandwagon. There have been a series of developments both in case of text and image generation, which has led to the current state of generative AI. In the, following section, we quickly do a time travel on these inventions.

- **1960-70:** n-gram models and Markov models were being used for language modelling. They continue to advance in the areas of Natural Language Processing for the subsequent two decades.
- **1980:** Autoencoders, though they were introduced for unsupervised learning, they provided a foundation for the encoding and decoding architecture of generative models. It is to be noted, much of autoencoder's glory was achieved later.
- **2003:** Yoshua Bengio and team created the first feed-forward neural network language model, capable of predicting the next word in a sequence. Predicting the next word is indeed a small but certain step towards generative AI

- **2014:** Generative Adversarial Networks (GANs) are introduced by Ian Goodfellow, revolutionizing the field by pitting two neural networks against each other, one generating content and the other discriminating between real and fake, leading to increasingly realistic outputs.
- **2014:** Variational Autoencoders - Deep latent variable models for generating new data points. The vanilla autoencoders have a deterministic behaviour, whereas the Variational Autoencoder brings an element of chance or probability, making their capability of generation, far more enriched.
- **2016:** PixelRNN/PixelCNN - Models for sequentially generating images pixel-by-pixel.
- **2020:** OpenAI unveils GPT-3, a large language model that exhibits remarkable fluency and coherence in text generation, sparking discussions about the potential and limitations of such powerful AI systems.
- **2022:** DALL-E 2, a diffusion model developed by OpenAI, takes image generation to new heights, producing stunningly realistic and creative visuals based on text prompts. Noise introduction as well as elimination is done in step-by-step manner in a diffusion model to generate a realistic image. The theory of Markov Chain is leveraged in the process.

  Figure 5.5 is an Image created using Generative AI.

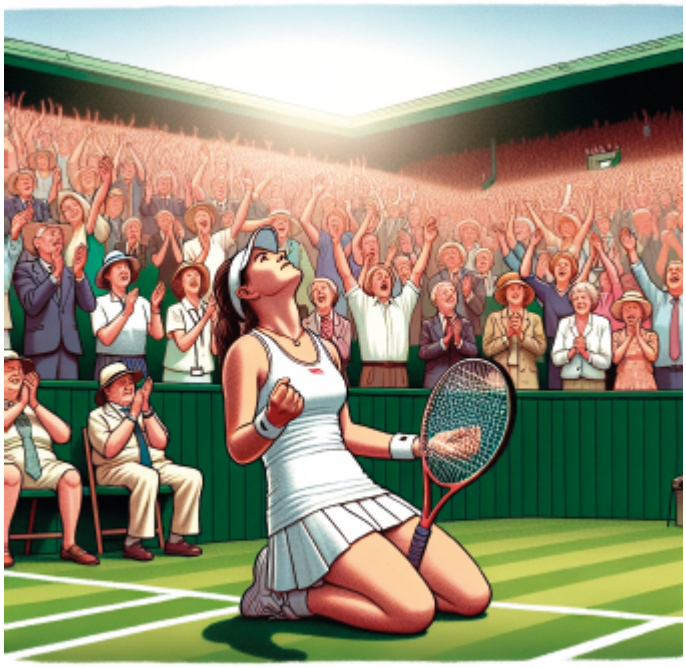


Figure 5.5 Image created by Generative AI



DO YOU KNOW

A deepfake is a synthetic media created using artificial intelligence (AI) and machine learning techniques to manipulate existing video or audio footage. This manipulation can range from subtle changes like altering someone's facial expressions to creating entirely new and realistic content featuring people who never actually said or did what is shown. In March 2023, a Canadian couple fell victim to a deepfake call scam, losing $21,000. The scam involved a fraudulent call where the perpetrator utilized technology to mimic the voice of the couple's son. The imposter falsely claimed to be in legal trouble, urgently seeking financial assistance for bail.

5.2.2 Synopsis of the Important Models

N-gram Models

N-gram models analyse sequences of words (n words at a time) to guess what's likely to come next. Like detectives gathering clues, they count how often words appear together, building a statistical map of language patterns. Unigrams consider single words, bigrams focus on pairs and trigrams analyse three-word sequences. The bigger the *n*, the more context they use, but also the trickier it gets to find enough examples in real text. Though simpler models, n-grams played a crucial role in early language processing and laid the groundwork for more powerful AI approaches.

- **Unigram:** Predicting the next word, without the context.
- **Bi-Gram:** Predicting the next word based on the last word
- **Tri-gram:** Predicting the next word, based on the last two words

Figure 5.6 depicts a lot of things.

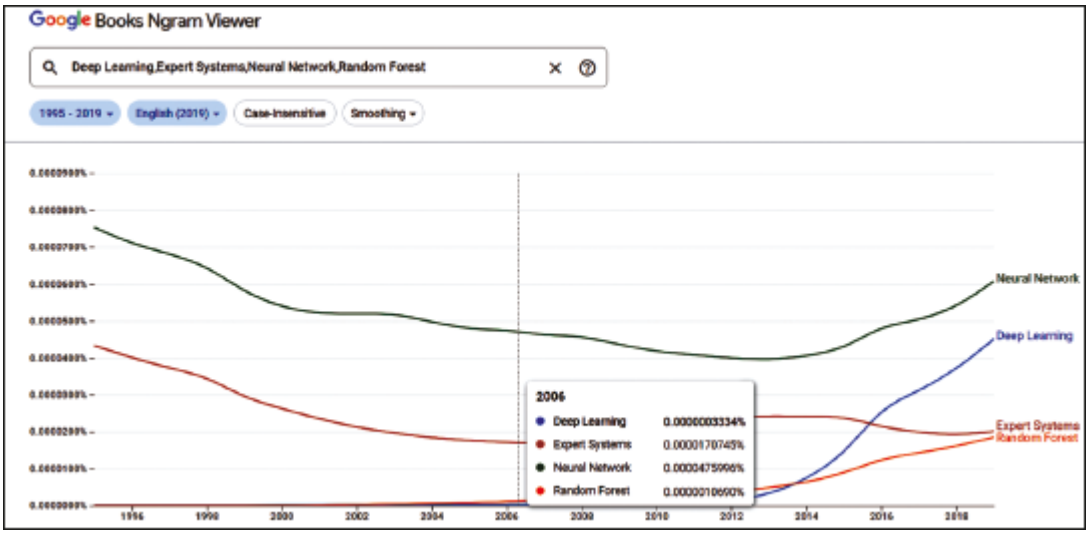


Figure 5.6 Google Books N-gram view of Deep Learning, Neural Network, Random Forest and Expert Systems

Google Books Ngram view allows us to see how these probabilities changed over the years. We find the cross-over pattern between deep learning (Part of Inductive AI) and Expert Systems (Part of deductive AI) really fascinating and revealing.

Auto-Encoders

- Autoencoders are a type of artificial neural network used for unsupervised learning and dimensionality reduction.
- They compress the input data into a lower-dimensional latent space representation and then reconstruct the output from this representation. For example, input the numbers 2, 6, 10, 14, 18, 22, 26 and the output is 30. So, there are 7 input variables. However, if we observe closely, the inputs are actually coming from an arithmetic progression, 2 is the first term and 4 is the common difference. Hence, rather than using the 7 inputs, we can use three inputs which are the first term, common difference and position of the term in the sequence.

$$F\ (2,\ 6,\ 10,\ 14,\ 18,\ 22,\ 26) \rightarrow 30$$

$$G\ (2,4,8) \rightarrow 28$$

As you see F is a function which works on the original input and G works on the first term, common difference and position of the term. These three numbers combined can be thought of as the latent representation. Now, neither in every problem, it is possible to find the la-

tent representation, nor in those cases the latent representation will be easily comprehensible as shown in this example.

- An autoencoder consists of an encoder that maps the input to a latent code, and a decoder that reconstructs the input from the code (**Fig. 5.7**).

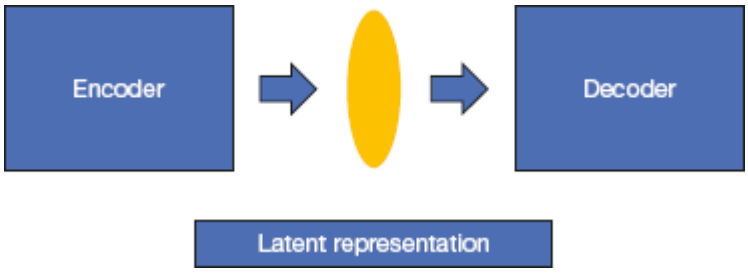


Figure 5.7 Block Diagram of Auto-Encoder

- The aim is to minimize the difference between the input and reconstructed output by optimizing the model parameters.
- Now, this latent space can be sampled and inputs can be given to the decoder, which will generate new points.

Generative Adversarial Networks (GANs)

The working of GANs is really interesting.

- Imagine two artistic rivals, constantly pushing each other to new heights. That's the essence of Generative Adversarial Networks (GANs)
  - **Generator:** An artist, trying to create realistic paintings (e.g., portraits, landscapes).
  - **Discriminator:** A critic, trying to spot fake paintings among real ones.
- The generator keeps improving its forgery skills, generating ever-more convincing artworks.
- The discriminator sharpens its detective skills, getting better at identifying fakes.



DID YOU KNOW

There is an incredibly cool thing called as Style Transfer. Let's say, we have a regular photo of a bustling city street and we want to render it Van Gogh's iconic swirling, streaky brushstroke style. This is totally possible using a popular GAN model named as StyleGAN. StyleGAN is a novel generative adversarial network (GAN) architecture for high-quality image generation developed by NVIDIA researchers in 2018–2019. It incorporates ideas from style transfer to give more control over image synthesis compared to regular GANs. It provides fine-grained control over image generation through injections of style vectors. The generator comprises a mapping network and synthesis network to disentangle style and noise. The mapping network transforms the latent code into an intermediate style representation. The synthesis network then generates the image from style and noise.

Next important set of models are the transformer models and the neural-based embedding models for text generation. But we postpone the discussion to the next section, where we discuss about ChatGPT and Prompt Engineering.

Diffusion Model

Diffusion models are a class of generative models that learn to reverse a multistep diffusion process in order to generate high-quality samples.

- They start with a real data sample, like an image and inject noise into it over multiple diffusion steps to obtain a noisy version.
- A neural network is then trained to take this noisy sample and predict how to remove some noise and reverse one step of the diffusion.
- By repeating this denoising process over many iterations, the model is able to remove all the injected noise and produce a realistic generated sample.
- Diffusion models are trained by optimizing the noise removal reconstruction at each step to mimic the data distribution.

- Key benefits are the ability to generate sharp, high-resolution outputs and excellent sample diversity.
- Recent variants like DDPM and GLIDE have shown impressive image and audio generation capabilities.
- The gradual denoising helps avoid spikes and artefacts common in other generative models.
- Diffusion models offer a noise-based perspective to generative modelling complementary to GANs and autoregressive models.

In summary, diffusion models take a stochastic approach to generation by training neural networks to model incremental noise removal from data. This yields highly realistic and diverse outputs across modalities.

5.3 CHATGPT AND PROMPT ENGINEERING

5.3.1 ChatGPT Context Setting

Let's quickly recap the development happening in NLP. There is no denying that machine-to-machine translation is one of the most complex tasks. This is a sequence-to-sequence task. i.e., input is a sequence and output is also a sequence. Regular neural network fails on sequence tasks. RNN and LSTM models came to the rescue. Here, the output is a function of the previous step output as well as the current state input.

However, in the real world when an interpreter tries to convert a sentence from one language to the other, he probably first listens to the entire sentence, understands the meaning and then states that in another language. So, it's not a word-by-word translation, rather arriving at an understanding and then restating it in the target language. Well, this is the intuition behind the encoder decoder network. As expected, this gave much better result than that of RNN and LSTM.

The next revolution in NLP came in terms of transformers. They used the encoder, decoder block and augmented it with attention. From an intuition perspective, this was more flexible than encoder, decoder. It's like we are not having the step of understanding the sentence fully and then regenerating, rather the transformer can look at the sentence in the source language in its entirety in the same instance while generating a

word in the target language and it pays attention to only the source language words that are relevant. One other advantage of transformers over encoder decoder is they can be trained in parallel.

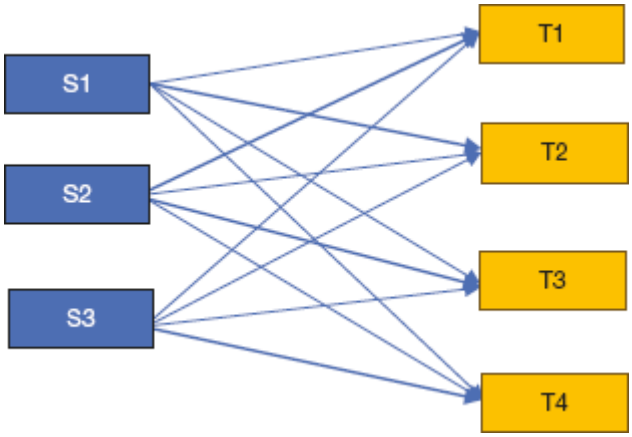


Figure 5.8 Attention Mechanism in Machine Translation

In **Fig. 5.8**, S1, S2 and S3 denote words in a sentence of the source language and T1, T2, T3 and T4 denote the words in the target language. There is all possible interconnection between each source and target pair of words. The thick lines indicate that while generating T4, attention needs to be made especially to S3. Such an interconnection layer, which allows to give specific weight to the inputs is called an attention layer.

In a more generic extension, there can be multiple layers of attention. So, one layer may look at the actual words, one layer can look at the part of the speech of the word. There can be several ways of paying attention to a word. One may be looking at the syntax of the words, another may be looking at the semantics of the words, yet another may be looking at the tone of the words. This technique is called multi-head attention.

With the advent of transformers, there was indeed a revolution in the way NLP tasks were performed earlier. While some transformers retained the encoder-decoder architecture, some altered these basic blocks.

The models that emerged as leaders initially were BERT (Bi-directional encoder representation) from Google and GPT (Generative Pre-Trained model) from Open AI. BERT's focus is more on understanding of language. As the name suggests, this focuses on the encoder part. Whereas GPT had its focus on the generative side i.e., the decoding part.

Apart from attention and parallel training, one other important characteristic of these transformers is pre-training. In terms of the number of parameters, amount of data, or CPU power these are gigantic beasts. Essentially, they are at first trained on large amounts of text mostly publicly available from Wikipedia or similar sources, even before they are given any language tasks to perform.

Before we move to elaborate on how the transformers are playing an important role, you may want to revisit, the important text processing tasks, that were discussed in **Chapter 2**.

Let's head to ChatGPT Now.

5.3.2 ChatGPT

ChatGPT is a super successful transformer from OpenAI. Following are the different versions of GPT.

- **GPT-2 (Generative Pre-trained Transformer 2):** Released in 2019, GPT-2 was a significant advancement with 1.5 billion parameters. It demonstrated powerful language generation capabilities, raising concerns about potential misuse.
- **GPT-3 (Generative Pre-trained Transformer 3):** Released in June 2020, GPT-3 is one of the largest language models, featuring 175 billion parameters. It excels in various natural language processing tasks and showcases remarkable text generation abilities.
- **ChatGPT 3.5:** Released in May 2022, GPT 3.5 brought many significant improvements and now GPT 4.0 is supposed to have close to 3 trillion parameters.

Let's quickly explore, some of the basic features of ChatGPT. The ensuing discussion is done in a generalized manner, so this will hold for Google BARD or Claude or any other similar tool.

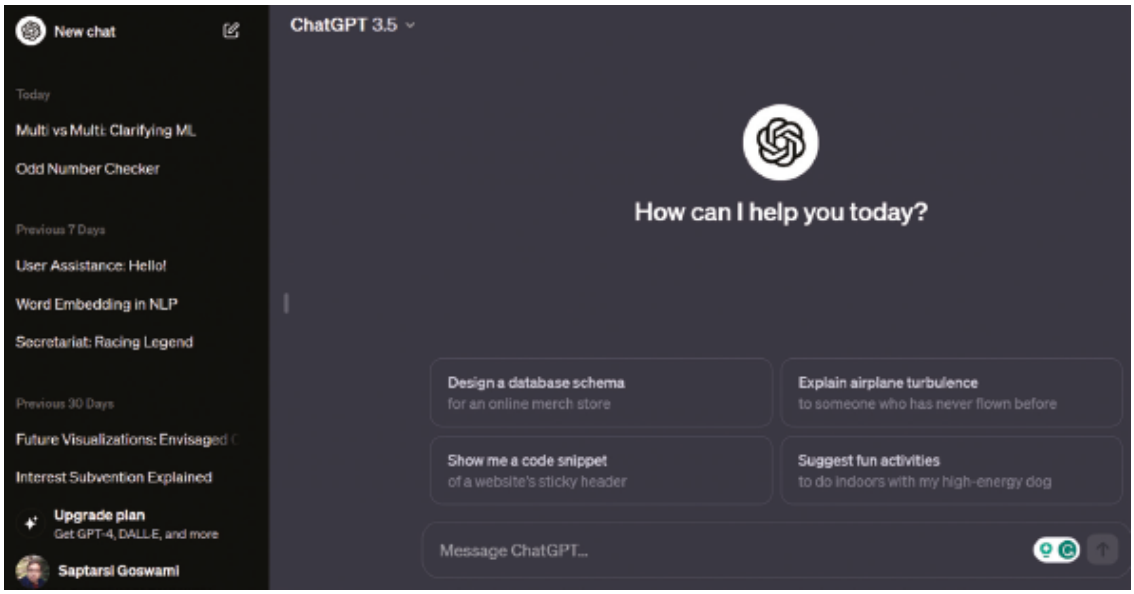


Figure 5.9 OpenAI ChatGPT Home Screen

If we look closely at **Fig. 5.9**:

- In the message part, you can ask a question and the question histories are saved.
- The question you ask is called a prompt. In the next section, we are going to read more about prompts (**Fig. 5.10**).
- These are probabilistic models; hence the result may be different to even the same question asked
- There will be an option to edit the question and resubmit.
- BARD can take input as audio as well as Images

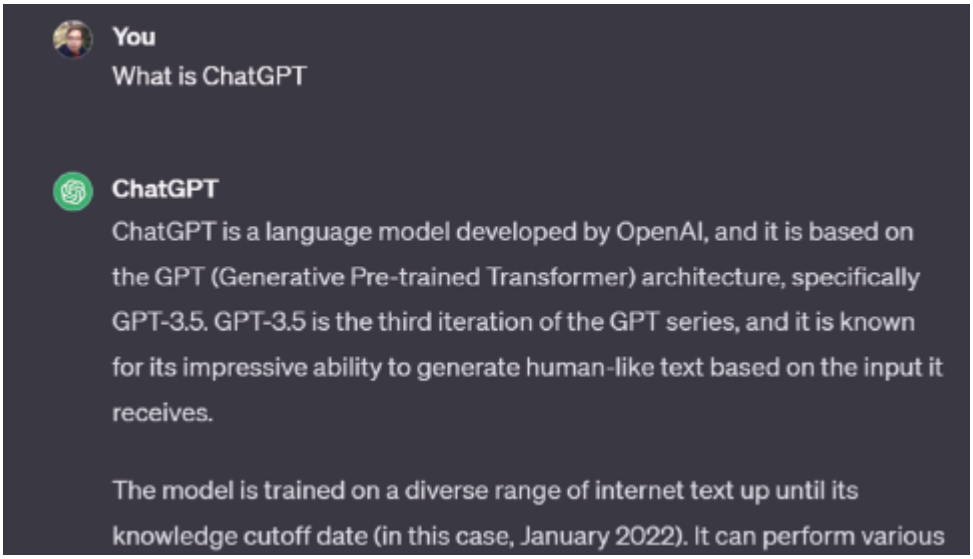


Figure 5.10 Prompt to ChatGPT and the response

- ChatGPT supports 90 + languages. If you wish to get output in a particular language, you will have to give the input also in that.
- After the response is generated, there are some options for liking, disliking and regenerating.
- Here's how, ChatGPT is continuously learning from humans, immediately it can know the feedback. In some cases, if you say, you dislike the response, it will go one step ahead and give another response and ask for your feedback again.

Below is a screenshot (**Fig. 5.11**) of when BARD was asked a question in Bengali.

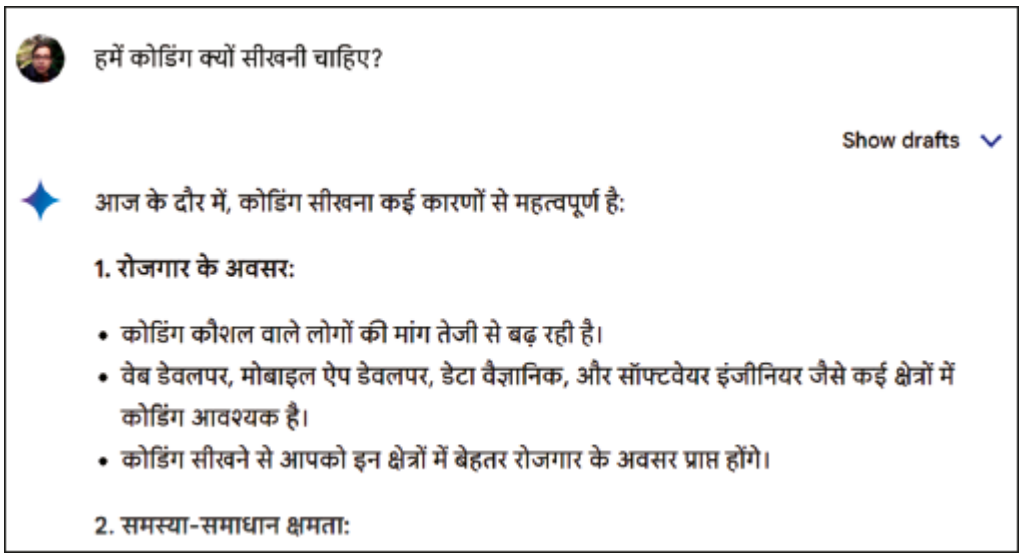


Figure 5.11 BARD now Gemini answering in Bengali

In the below instance, Google BARD can be found to be actually describing a diagram (**Fig. 5.12**)

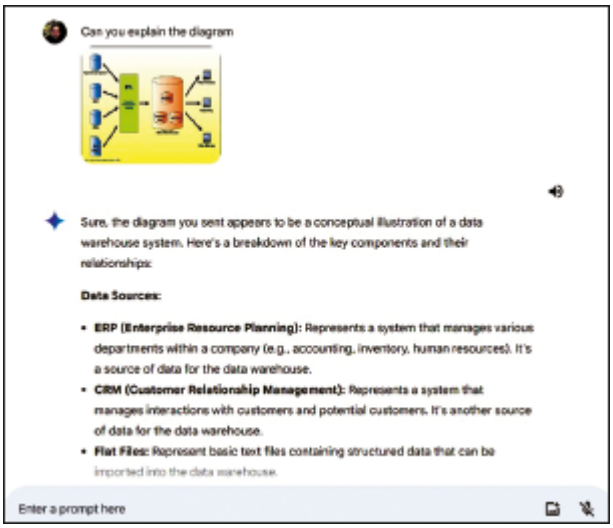


Figure 5.12 BARD now Gemini describing a diagram

5.3.3 Prompt Engineering

The questions that we ask to ChatGPT are called prompts. While ChatGPT can do exceedingly well, in answering it also needs well-formed questions, which are clearly formulated has the context associated with it. This art of asking the right question or formulating the right prompt is called prompt engineering. It is being touted.

This can be very clearly observed from the following Google Trends Graph (**Fig. 5.13**).

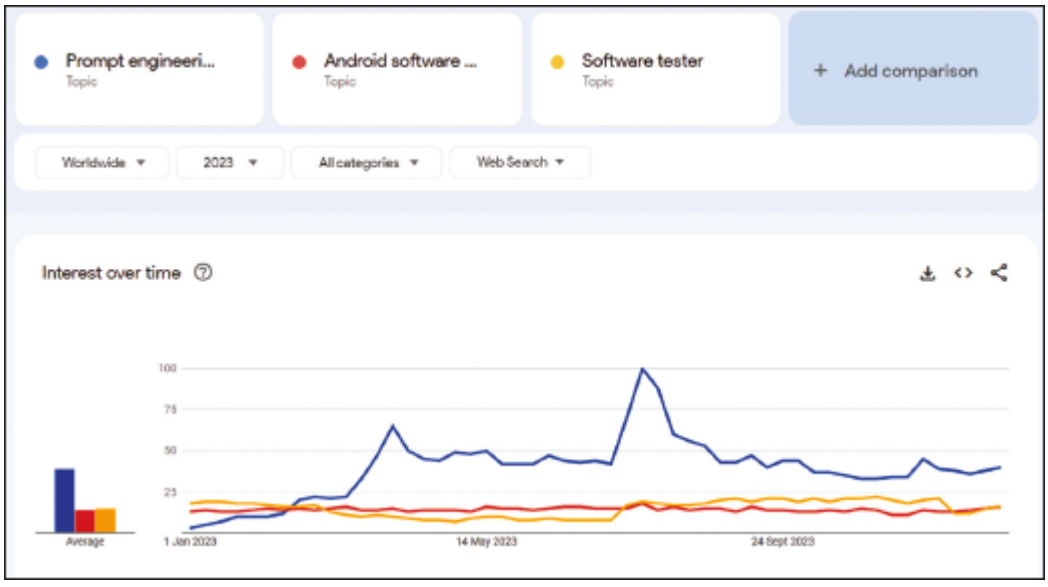


Figure 5.13 Comparing Prompt Engineering, Android, Software Tester over time using Google Trends

Prompt Engineering is searched more than, traditional software skills like Android Development or Software Testing.

In the below diagram, we can see, the affinity of different countries to prompt engineering in the world map (**Fig. 5.14**).

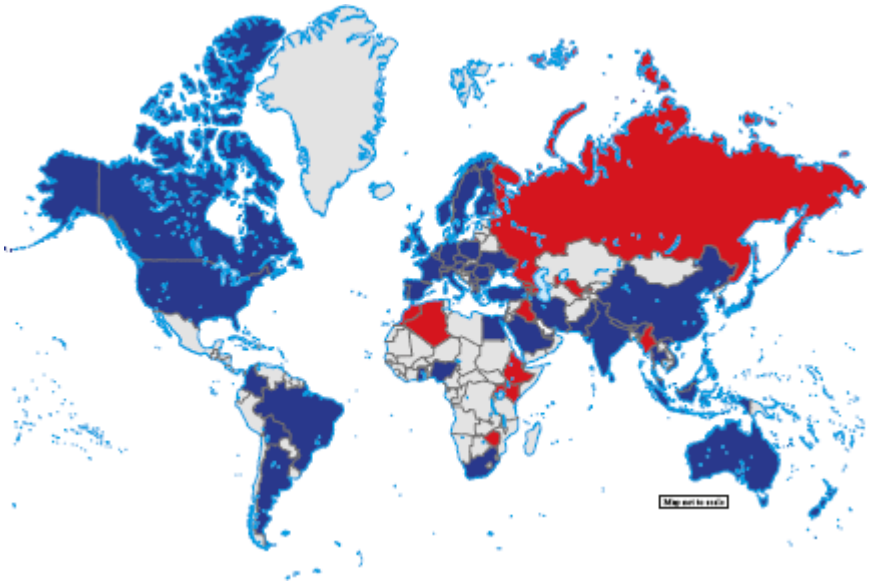


Figure 5.14 Comparing Prompt Engineering, Android, Software Tester over time using Google Trends

As it can be observed the entire world is curious about prompt engineering (Regions marked in darker shade), except areas like Russia and some low volume search regions (Which are greyed out).

The key aspects of prompt engineering include:

- **Task Framing:** Expressing the desired task clearly in natural language at the start of the prompt.
  Examples:
  - **Ok Prompt:** What is AI?
  - **Better Prompt:** Give the definition of AI with a few examples on application of AI
- Control the length of the output - You can also limit the number of words, character in the output.
  Examples:
  - **Ok Prompt:** What is GDP?
  - **Better Prompt:** What is GDP, explain in 100 words.

- **Better Prompt:** Please explain GDP with 3-4 bullet points

- Control the tone or nature of the output - You can specify the tone or manner in which you want the output.

  Examples:

  - **Ok Prompt:** What is Industry 4.0?
  - **Better Prompt (Fig. 5.15):**

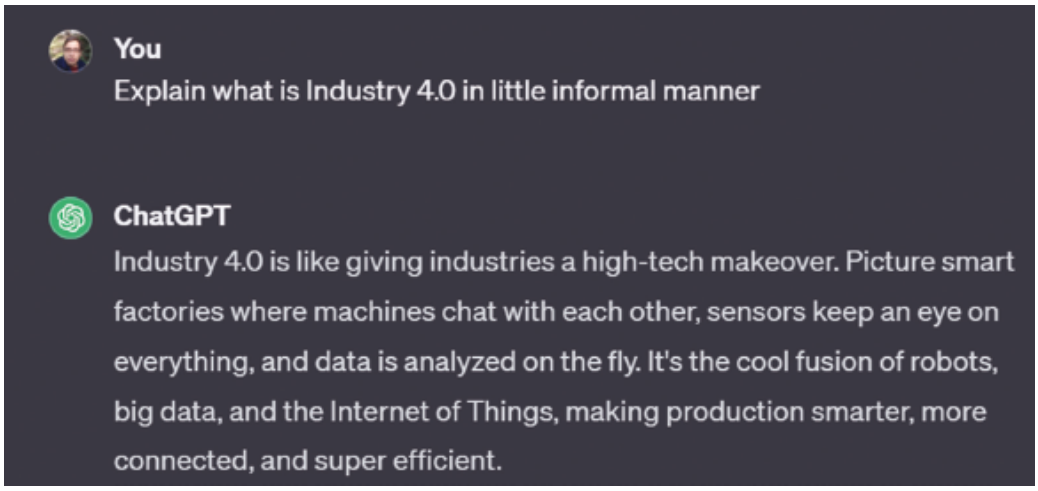


Figure 5.15 Better Prompts

  - **Better Prompt (Fig. 5.16):**

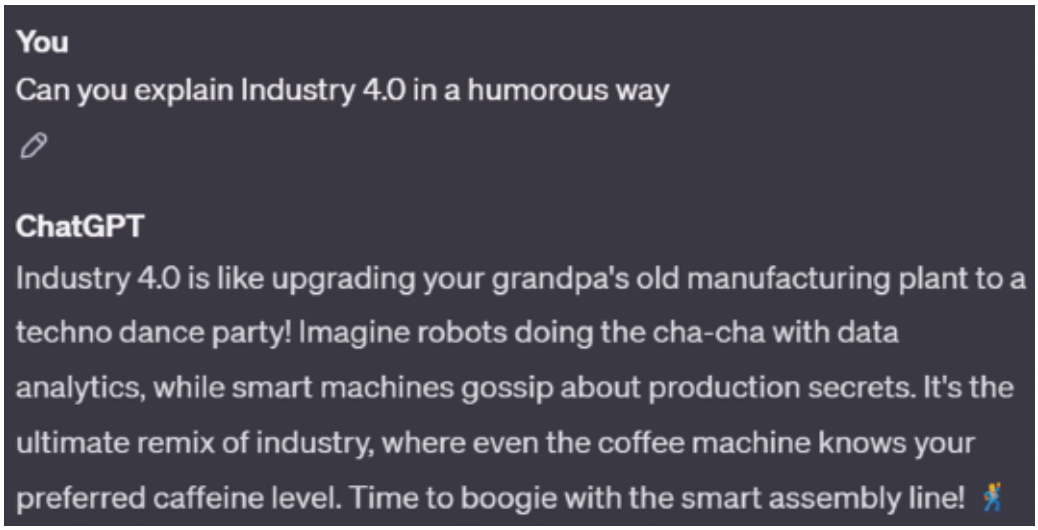


Figure 5.16 Better Prompts

- **Examples:** Providing a few demonstration examples that illustrate the input and expected output. This can be seen in the below diagram

(**Fig. 5.17**).



Figure 5.17 Prompt with specific output format

- One excellent technique here is to 'Explain to a 4th Grader' technique (**Fig. 5.18**). If there is a complicated concept, which you want it to be stated in a simple manner comprehensible to a 10-year-old, this can be used.

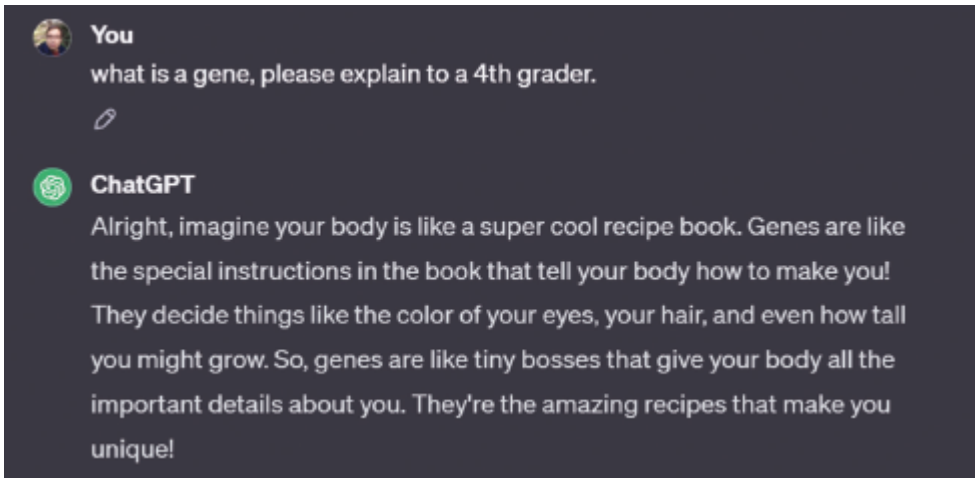


Figure 5.17 Fig.5.16 Explain to 4th Grader Technique

- Another useful prompt technique is role play. You can ask to role play as a fitness coach, a nutrition expert, a motivational speaker and so

forth (**Fig. 5.19**).

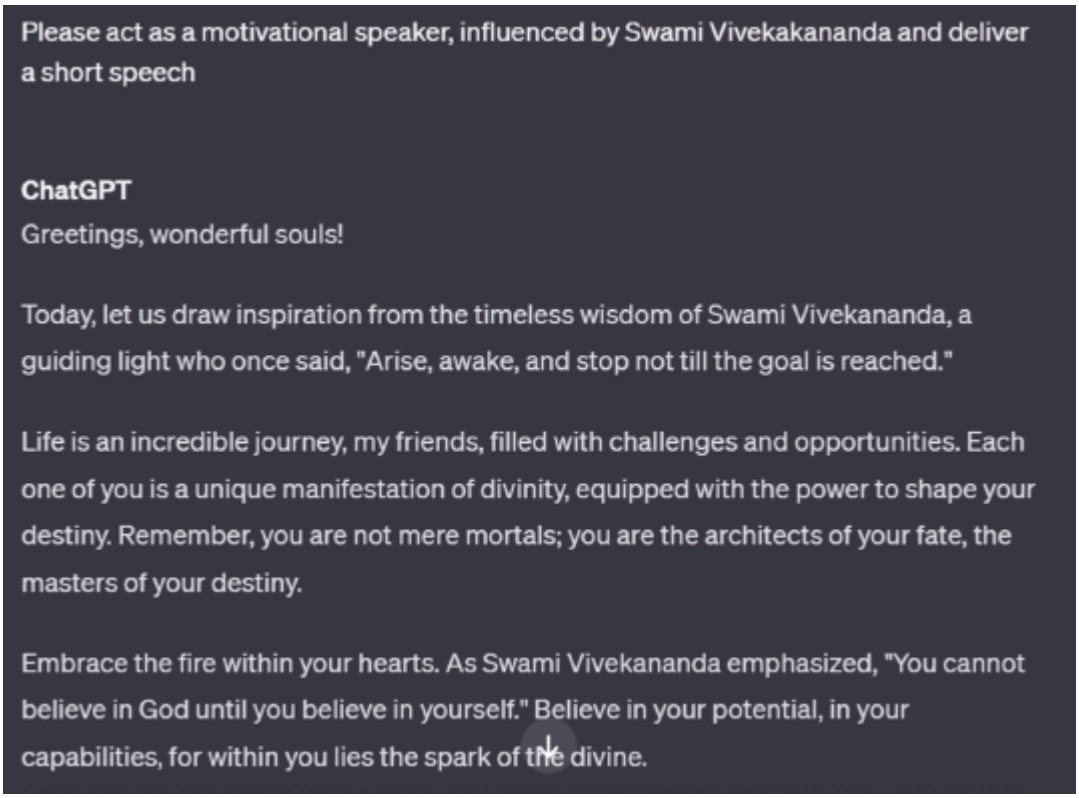


Figure 5.19 Prompt Engineering – Role Play or 'Act as a'

You can be really imaginative and creative in designing the prompts, which cater to your particular needs.

- **Prompt Priming:** Prompt Priming is a technique, where you take a very step-by-step approach. Rather than a direct question. Let's say, we want to get a delicious chicken recipe, rather than asking it directly, we can start by saying 'We are having some guests today, who are Punjabi, what kind of food will be good for them'. The very first option, that it gives is:
  'Butter Chicken (Murgh Makhani): A creamy and indulgent tomato-based curry with tender chicken pieces.'
  Then, you can say about other dishes, like you are going to eat with roti or rice, if you have already some gravy items and then you may specify that.
- **Chain of Thoughts (CoT):** In 2022, researchers at Google including Wei proposed an approach called Chain-of-Thought (CoT) prompting to improve large language models' performance on certain complex tasks. This method has the AI break down its thought process into intermediate reasoning steps.

Below is an example of CoT for solving a math word problem:
Demonstration: "A train travels 100 km in 2 hours. What is its speed?"
Answer: '50 km/h. (Explanation: distance/time = 100 km/2 h = 50 km/h).'
Prompt: 'A car travels 150 km in 3 hours. What is its speed?' Follow the steps from the demonstration to solve this problem.

- **Few Shots Learning:** In prompt engineering, few-shot learning is a technique where we leverage the capabilities of large language models (LLMs) to perform specific tasks by providing them with a limited number of examples (typically 2-10). It essentially teaches the LLM by demonstration, helping it grasp the desired output format and style without extensive training on large datasets.
  - Many of the tasks, ChatGPT or other agents can do, without any example. This is called as Zero-Shot learning.
  - When one example is given, it is often called One-Shot Learning

While the above discussion is focused on text, the below table gives a synopsis of tasks related to other type of contents and the tools associated.

Type of Content	Type of Task	Tool
Image	Image Generation from Text	Midjourney, Dall-E 2
Audio	Music Generation	MuseNet
Video	Generate Facial Expressions	FaceSwap
Audio	Voice Cloning	Resemble
Image	Photorealistic Images	Artbreeder

Following point regarding the examples and tools need to be noted

- This table and the examples using ChatGPT/BARD (Gemini) is not exhaustive and represents some of the popular tools available.
- Tool availability and capabilities may change over time.
- Some tools may have limitations or require paid subscriptions.

5.4 EMERGING TRENDS AND FUTURE DIRECTIONS IN AI

Some of the important trends and directions which are going to bring disruptive and transformative changes are as follows:

Technological Advances

- **Advances in Deep Learning:** Such techniques include transformers, diffusion models, and foundation models which are essentially larger models with billions of parameters, trained on massive datasets. They are capable of multimodal AI that leverages vision, language and robotics. Increased integration of modalities.
- **Distributed Systems and Federated Learning:** Federated learning is an approach to machine learning where the model is trained across multiple decentralized devices or servers holding local data samples. Instead of sending raw data to a central server for training, the model is sent to individual devices, which compute updates based on their local data. These updates are then aggregated to improve the global model.
- **Causal AI and Machine Reasoning:** Causal AI is a subfield of AI that focuses on understanding and reasoning about cause-and-effect relationships. Unlike traditional AI which excels at prediction based on patterns, Causal AI aims to explain why things happen and how interventions can influence outcomes. AI chips and specialized hardware. New processors optimized for AI workloads.
- **AI Chips and Specialized Hardware:** GPUs have become a standard for training neural networks given their parallel processing power. Nvidia GPUs are widely used for deep learning. Competition is emerging from dedicated AI accelerators like TPUs (Tensor Processing Units) developed by Google specifically for machine learning workloads.

Responsible AI Practices

- With AI being an intertwined part of everyday life, there is an increased focus on AI safety, ethics and alignment. Hence, the key themes looking forward are increasing scale and computing, safety and ethics, making AI accessible and beneficial, and gaining a deeper understanding of reasoning, causality and the world. This we have previously discussed in **Chapter 4**.

AI with its Expanded Reach

- Some other trends are personalization and customization of AI where the models are tailored for individual users and use cases. AI is being used increasingly in science, automating and augmenting scientific discovery in materials, medicine, climate etc. There is emphasis on democratization of AI by easier access through pre-trained models, libraries, tools, etc.



DO YOU KNOW

CRFM, a Stanford HAI initiative, brings together faculty, students and researchers from over 10 departments to advance the science and responsible development of foundational language models. With a focus on ethics and societal impact, CRFM at Stanford HAI is dedicated to ensuring AI's powerful language models are used responsibly and contribute positively to society. The more can be found at **https://crfm.stanford.edu/**

5.5 AI AND SOCIAL INEQUALITY

AI in one hand can increase inequality again it has also the potential of mitigating the risk.

AI and Social Equality

AI can potentially automate many tasks. Automation may displace jobs, which may hit the lower income roles hardest.

Loss of Jobs

- **Routine and Repetitive Tasks:** Jobs involving repetitive manual tasks, such as assembly line work or data entry, may see a decrease in demand as AI and robotic automation become more advanced.
- **Customer Service:** Basic customer service roles, especially those involving routine inquiries or transactions, may be automated using AI-powered chatbots or virtual assistants.

- **Data Analysis:** Jobs that primarily involve data analysis and reporting could be affected, as AI algorithms can process and analyse large datasets more quickly and efficiently than humans.
- **Driving and Transportation:** With advancements in autonomous vehicles, jobs related to driving, such as taxi drivers and truck drivers, may see changes. Automated delivery services could also impact certain logistics jobs.
- **Manufacturing and Assembly:** AI and robotics can be used for tasks like manufacturing and assembly, potentially reducing the demand for manual labour in certain industries.
- **Routine Healthcare Tasks:** Jobs involving routine healthcare tasks, such as medical data entry and some diagnostic processes, could be affected as AI systems are developed for these functions.
- **Financial and Accounting Tasks:** Some financial and accounting tasks, including basic bookkeeping and data entry, could be automated using AI systems.

AI tools and services if are available to only a privileged few, that can create a divide between these groups. Use of AI needs access to computing devices, internet and sometimes subscription to services. These facilities are not typically available to all. Those with limited access to education and resources may lack the skills needed to engage with and benefit from AI technologies, creating a knowledge as well as skills divide. Some other issues like Bias, lack of diversity can produce results which are not best for equality. These issues are also elaborated in detail.

How AI Could Promote Equality

AI technologies can improve access to education, healthcare, and information, bridging gaps and providing opportunities to underserved communities. There are many AI solutions for children with special needs.

Khan Academy Kids, this free platform utilizes an AI-powered learning engine that personalizes the learning experience for each child. Language-learning apps like Duolingo use AI to give feedback and support to new language learners from underserved communities. Inclusive design, testing and auditing of AI systems by diverse teams improve social awareness.

5.6 SUMMARY

- AI aids in generating hypotheses, virtual prototyping, data collection, and analysis in various scientific activities.
- AI finds application in Particle Physics, Astro Physics and Space Experimental Chemistry, Biology, Climate Modelling and many more
- Generative AI is a specialized branch of creating new content like text, images, video and music.
- **Timeline of developments:** N-gram models, autoencoders, GANs, Variational Autoencoders, PixelRNN/PixelCNN and recent breakthroughs like GPT-3 and DALL-E 2.
- N-gram Models analyse word sequences, with unigrams, bigrams, and trigrams predicting the next word based on context. Autoencoders are neural networks for unsupervised learning and dimensionality reduction, compressing input data into a lower-dimensional latent space. Generative Adversarial Networks (GANs) use Artist (Generator) *vs.* Critic (Discriminator) concept, creating ever-more convincing artworks.
- Diffusion Model is a class of generative models for high-quality sample generation. Injects noise into real data over multiple steps, denoising to generate realistic samples.
- Transformers use attention mechanisms and parallel training for more flexible and efficient processing. ChatGPT introduced by OpenAI is based on transformer technology with outstanding capability in standard NLP tasks include spam detection, NER recognition, sentiment analysis, machine translation, text summarization, Q&A, and text generation.
- Prompt Engineering is a technique of crafting well-formed and contextually clear questions for ChatGPT, known as prompts.
- Some of the techniques to improve the prompts are task framing, controlling output length and tone, Prompt Priming, ‘Act as’, ‘Explain to’, etc.
- Advanced Prompt Engineering Techniques include Chain of Thoughts where LLMs are guided to provide a step-by-step reasoning for solving a mathematical or logical problem.
- Few Shots learning, on the other hand, provides the LLMs with some examples, which are to be followed in producing the output.

- Some of the technology advances of course include technology advances in deep learning, federated learning, specialized hardware like GPU, TPU. AI is getting democratized and is available for everyone's use. However, with the widespread use of AI, there is an increasing thrust on responsible AI.
- AI posed challenges like loss of Jobs, Access Divide, Bias and Lack of Diversity, which influence decisions. Some of the opportunities it presents are improved access and inclusive design.

5.7 PRACTICE EXERCISES

5.7.1 Subjective Question (10 Marks)

1. How does AI contribute to experimentation in various scientific disciplines and what specific activities can AI assist in?

2. How can AI be applied in environmental science, specifically in climate modelling, air and water quality monitoring, waste management and resource conservation?

3. Define Generative AI and its role in generating new content, highlighting the importance of coherence and passing the Turing Test.

4. Differentiate between discriminative models and generative models in machine learning, providing examples of each.

5. Outline the chronology of developments in generative AI, emphasizing key milestones and breakthroughs from 1960 to 2022.

6. What are the significant contributions of OpenAI in the field of generative AI and how have other major organizations, such as Meta and Google, joined the advancements?

7. Explain the functionality and purpose of n-gram models in language modelling, describing their role in early language processing.

8. Elaborate on the concept of Generative Adversarial Networks (GANs). Highlight the roles of the generator and discriminator in the generation of realistic content.

9. How do diffusion models work in generative modelling and what benefits do they offer in terms of image and audio generation?

10. Describe the significance of transformer models and neural-based embedding models for text generation in the context of AI.

11. Elaborate on the key aspects of prompt engineering, focusing on the importance of task framing, controlling the length of output and controlling the tone or nature of the output.

12. Explain the concept of prompt priming and how it differs from direct questioning. Describe the same with a suitable example.

13. What is the Chain-of-Thought (CoT) prompting approach and how does it contribute to improving large language models' performance on complex tasks?

14. Describe the few-shot learning technique in prompt engineering, emphasizing its role in teaching large language models specific tasks with a limited number of examples.

15. Explore the emerging trends and future directions in AI. Focus on technological advances, responsible AI practices and AI's expanded reach in various fields.

5.7.2 Subjective Question (5 Marks)

1. Explain the role of AI algorithms in identifying particles at the Large Hadron Collider (LHC) and the challenges it addresses.

2. How is AI utilized in astrophysics and space research, particularly in analyzing images of galaxies and simulating celestial phenomena?

3. In what ways does AI enhance efficiency and analysis in experimental chemistry, and what benefits does it bring to the discovery of novel compounds?

4. Describe the use cases of AI in biology, including predicting protein structures and tracking neural activity in the brain.

5. What is Prompt Engineering, and why is it considered a crucial skill for unlocking the true potential of ChatGPT?

6. How can the 'Explain to a 4th Grader' technique be effectively employed in prompt engineering, especially when dealing with complex concepts?

7. Discuss the role-play technique in prompt engineering and provide examples of scenarios where role-playing can be beneficial.

5.7.3 Objective Question (2 Marks)

1. Why GAN is called an adversarial network?

2. What are the different n-gram models?

3. What are some of the specialized hardware for AI?

4. What is Prompt Engineering?

5. What are the two benefits of using well-formed prompts?

6. Explain Aspect Based Sentiment Analysis (ABSA) with a suitable example.

7. Briefly describe, what you understand by machine translation.

8. What is causal AI?

5.7.4 Multiple-choice Questions (1 Mark)

1. How would you define generative AI in simple terms?

(a) A type of AI used for deep learning

(b) AI focused on discrimination tasks

(c) AI that generates new content

(d) AI specializing in computer vision

2. What is the key characteristic of discriminative models in machine learning?

(a) They focus on generating new data

(b) They predict outputs given inputs

(c) They work only with images

(d) They ignore input data

3. Which model introduced in 2014 revolutionized the field by using two neural networks against each other?

(a) PixelRNN

(b) Variational Autoencoders

(c) Generative Adversarial Networks (GANs)

(d) Autoencoders

4. What is the primary capability of Diffusion Models in generative modelling?

(a) Predicting climate patterns

(b) Denoising and generating realistic samples

(c) Enhancing data analysis

(d) Simulating astrophysical phenomena

5. What is prompt engineering?

(a) Traditional software development

(b) Art of asking questions to ChatGPT

(c) Building physical structures

(d) Writing computer code

6. In prompt engineering, what does ‘Task Framing’ involve?

(a) Crafting a picture for the wall

(b) Clearly expressing desired tasks in natural language

(c) Creating a new programming language

(d) Designing a piece of furniture

7. What is the purpose of the ‘Explain to a 4th Grader’ technique in prompt engineering?

(a) Simplifying complex concepts for better understanding

(b) Training 4th graders in advanced topics

(c) Developing new educational materials

(d) Testing knowledge of 4th-grade students

8. What is Chain-of-Thought (CoT) prompting in AI?

(a) Sequentially chaining tasks in AI development

(b) A method for improving large language models’ performance

(c) Creating a chain of AI-powered devices

(d) Training AI models with thought processes

9. What is few-shot learning in prompt engineering?

(a) Learning with very few examples

(b) Intensive training with numerous examples

(c) Simultaneous learning of multiple tasks

(d) Learning without any examples

5.8 ANSWER KEYS

5.8.1 Multiple-choice Questions

1. (c)

2. (b)

3. (c)

4. (b)

5. (b)

6. (b)

7. (a)

8. (b)

9. (a)